Assignment: Behaviour Trees - REVISED VERSION

IMPORTANT: For this assignment, be sure you are using version 0.4.1 (or greater) of the AAPE environment. You will find the releases for each operating system, as well as the updated guide, in the "AAPE - Autonomous Agents Practice Environment" section of the Campus Virtual.

In this assignment, we will use Behaviour Trees to implement relatively complex behaviours within the AAPE platform. Our main characters will be the **Astronaut** and **CritterMantaRay** agents.

Testing Environment

- The primary test scene is named "**3-AAC**". This scene will be used to evaluate your agents (applying to all three scenarios described below).
- *Note:* You may use other scenes for additional testing if needed.

Implementation Scenarios

You are required to implement **three incremental scenarios**. The table below outlines the scoring for each successfully implemented scenario.

Scenario	Score	Comments
Alone	5-6-7	This is mandatory and necessary to pass the assignment. You can earn 5, 6, or 7 points by implementing this scenario.
Critters	+1	Implementing this scenario will increase your score by up to 1 point.
Collect-and-run	+2	Implementing this scenario will increase your score by up to 2 points.

<u>Scenario Alone</u>

Our Astronaut has been assigned to the DarkHole deep-space collection outpost.
Her mission is to collect a rare species of alien flower. Fortunately, she is alone at the outpost, so gathering the flowers should be straightforward.

The Astronaut can start from one of the Base outpost (SpawnAlpha, SpawnBeta, SpawnGamm, SpawnDelta). I recommend you use BaseAlpha for the initial tests because you will find flowers very near (NearBaseZone). Check AAPE-Guide-V0.4.1.pdf for the details about this scenario (spawn points, areas, etc). The number of alien flowers remains constant. Whenever the astronaut collects a flower, a new one will spawn randomly in the harvest zone after a short delay.

Behavior:

- **Wandering & Detection**:
  - The astronaut wanders the outpost searching for alien flowers (tag: `AlienFlower`).
  - When a flower is detected, she moves toward it and automatically collects it (added to her inventory).
- **Inventory Limit**:
  - She can carry only **two flowers** at a time.
  - When her inventory is full, she must return to the *Base Outpost* to unload (`action:leave,AlienFlower,2`).
- **Returning to Base**:
  - To return to the base outpost, she uses the **NavMesh system** (`action:walk_to,BaseAlpha`).
  - The Astronaut may also use instant teleportation (`action:teleport_to,BaseAlpha`) to speed up the process (since the return trip in this scenario is always safe).

<u>Scenario Critters</u>

We now move to a new collection outpost, situated on a small moon near Saturn. Recently, hostile life forms have been discovered here (`"type": "AAgentCritterMantaRay"`).

Assignment: Implement the behaviour of these life forms.

Behaviour Requirements:

1. **Default State (Roaming):**
   - The critter roams aimlessly, avoiding obstacles.
2. **Astronaut Detection:**
   - If the critter detects the astronaut, it **starts following her** and attempts to bite (touch the astronaut).
   - Biting the astronaut implies the astronaut is stunned for 5 seconds, and if she had flowers in her inventory, she will lose one (this behavior

is automatic and managed by AAPE, you don't have to program it in Python)
    - After biting the astronaut, the critter must **move away** and allow the astronaut to recover.
    - If it loses contact, it **resumes roaming**.

TIPS:

- If you want to try with multiple critters, use the Spawner.py tool and the two Spawn Areas ("HarvestZone" or "SmallHarvestZone").
- Use a manually controlled Astronaut to test your Critters.

Scenario Collect-and-run

Our Astronaut has arrived at the Soylent Green outpost to continue with her task of collecting alien flowers. However, this time is not so easy. We have the critters wandering around, and the Astronaut has to avoid them at all costs. A bite from a Critter means the Astronaut gets paralysed for 5 seconds; moreover, the Critter takes one of the flowers from the Astronaut's inventory.

Behaviour Requirements:

We want the **Astronaut** to replicate the same behavior demonstrated in the *"Alone"* scenario, but now with an added requirement: **active avoidance of the Critters**.

Your implementation will be assessed based on how successfully the autonomous Astronaut evades the Critters.

Naturally, we expect the Astronaut's behavior to improve compared to the 'Alone' scenario when facing the Critters. The scoring will be based on this demonstrated improvement.

In this scenario, you have to use the *walk_to* base mode, and try to avoid the Critters also when you are returning.

Some notes about the 3-ACC scenario

You will find a description of this scenario in the AAPE-Guide-V0.4.1.pdf with the spawn areas, spawn points and locations.

To adjust the number of AlienFlowers in each spawn area, you can modify the file "config.json" in the directory "StreamingAssets"[1], which can be found in your directory release (see in the footer the location for each specific release). The file has the form:

```
{
 "flowersInHarvestZone": 10,
 "flowersInSmallHarvestZone": 0,
 "flowersInNearBaseZone": 0
}
```

The default config.json file that comes with the release spawns 10 AlienFlowers in the HarvestZone area, plus 5 in the NearBaseZone. Modify the area parameters as needed for testing your agents. Be aware that the evaluation test will be performed by spawning AlienFlowers only in the "HarvestZone" and "SmallHarvestZone" areas.

Remeber that Locations are the tags that can be used with the "walk_to" and "teleport_to" actions ("teleport_to" can also use spawn points as target teleport positions). Use the exact name together with the desired action.

The locations are detected by the sensor of the Astronaut ({'name': 'BaseAlpha', 'tag': 'Location'...}).

Figure 1 shows the area around a container where the Astronaut must remain to interact with that container. If you *walk_to* or *teleport_to* the *BaseAlpha, BaseBeta, etc.* locations, you will automatically be inside the area. This area is detected by the sensor of the Astronaut ({'name': 'Container', 'tag': 'Container'...}).

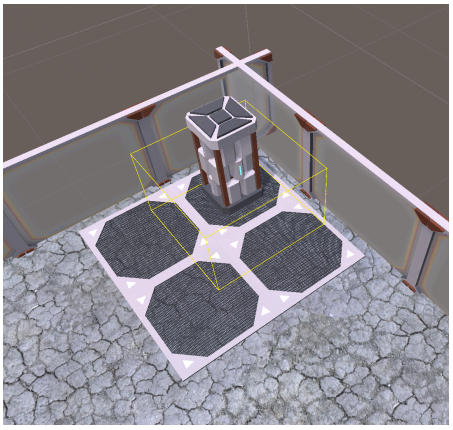


Figure 1

[1] Windows: AAPE_Data\StreamingAssets
Mac: Contents/Resources/Data/StreamingAssets
Linux: Linux-V0.3.0_Data/StreamingAssets

Finally, the **tags** of the different actors are the following:

AlienFlower Astronaut CritterMantaRay

The rest of the elements of the scenario are tagged as “Rock” (each one of the rock formations) and “Wall” (the walls that delimit the perimeter).

Provided code

An example behaviour tree, `BTRoam.py`, is provided. This is the same one we discussed in class. Keep in mind that this is just a reference example.

The file `Goals_BT_Basic.py` includes three example goals (DoNothing, ForwardStop, ForwardDist, and Turn), so you can try the behaviour tree in the BTRoam.py file. You should replace/extend this file, adding any other necessary goals.

`AAgent-Alpha.json`, `AAgent-Beta.json`, `AAgent-Delta.json`, `AAgent-Gamma.json`, are the configuration files for the **Astronauts** (the difference among them is just the color of the Astronaut), while `AAgent-Critter.json` is for the **Critter** agent. You may modify these files as needed or create new ones. Additionally, APackCritter.json and `APackAstroCritters.json` are example configuration files for the **Spawner** (`Spawner.pt`).

The AAgent_BT.py is the agent's main code. Initially, you only need to modify that file to register your goals and behaviour trees.

```
# Reference to the possible goals the agent can execute
self.goals = {
    "DoNothing": Goals_BT.DoNothing(self),
    "ForwardDist": Goals_BT.ForwardDist(self, -1, 5, 10),
    "Turn": Goals_BT.Turn(self),
}

# Reference to the possible behaviour trees the agent can execute
self.bts = {
    "BTRoam": BTRoam.BTRoam(self)
}
```

AAgent_BT code section where you have to register the goals and behaviour trees so they can be called from the AAPE interface (“Send message” button).

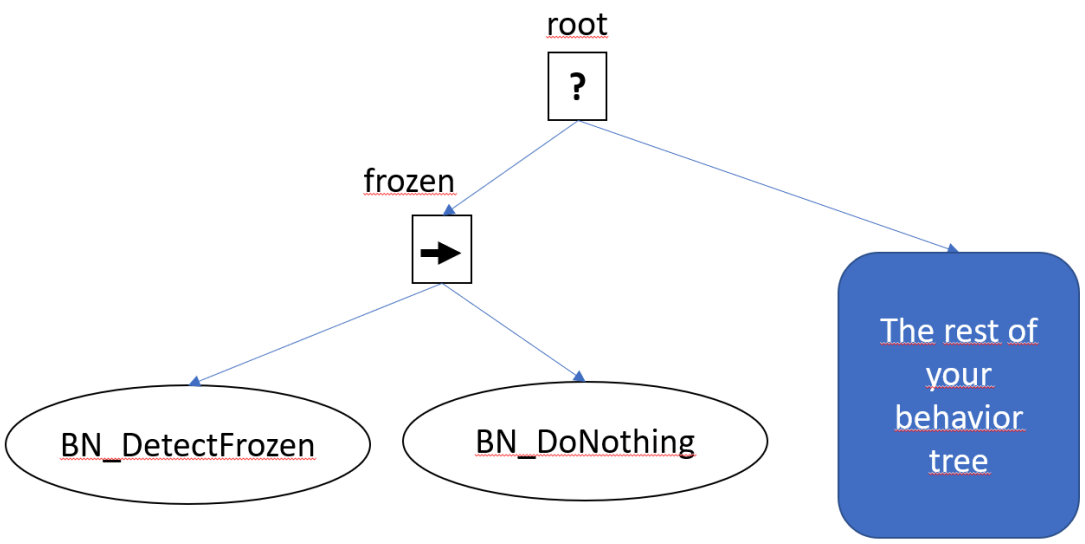
In this file, you can modify the AAgent class to add extra code if you decide to add new functionality that cannot be implemented as goals or a behaviour tree. This is allowed, but you have to mark your code clearly like this:

```
<current code in the AAgent class>

#------------------------------ CUSTOM CODE -------------
<your custom code>
#---------------------------------------------------------

<current code in the AAgent class>
```

<u>You cannot modify the provided functions or the current properties of the agent class. You are only allowed to add new elements.</u>

<u>IMPORTANT NOTE ABOUT THE IMPLEMENTATION OF THE BEHAVIOR TREE</u>

To manage correctly the period when the astronaut is stunned after a critter's bite, your behavior tree has to comply with the following structure:



Since the AAPE simulation blocks incoming messages to the astronaut while stunned, this structure ensures your behavior tree does not send actions to the simulation during that period. This prevents potential issues (such as unexpected behavior) when the astronaut resumes using the behavior tree after recovering.

BN_DetectFrozen should have this implementation:

```
class BN_DetectFrozen(pt.behaviour.Behaviour):
   def __init__(self, aagent):
       self.my_goal = None
       # print("Initializing BN_DetectInventoryFull")
       super(BN_DetectFrozen, self).__init__("BN_DetectFrozen")
       self.my_agent = aagent
       self.i_state = aagent.i_state

   def initialise(self):
       pass

   def update(self):
       if self.i_state.isFrozen:
           return pt.common.Status.SUCCESS
       return pt.common.Status.FAILURE

   def terminate(self, new_status: common.Status):
       pass
```

BN_DoNothing is the one you already have in the example.

The part of the behavior tree will be:

```
frozen = pt.composites.Sequence(name="Sequence_frozen", memory=True)
frozen.add_children([BN_DetectFrozen(aagent),BN_DoNothing(aagent)])
```

THE REST OF YOUR BEHAVIOR TREE DEFINITION

```
self.root = pt.composites.Selector(name="Selector_root", memory=False)
self.root.add_children([frozen,
                        THE_ROOT_COMPOSITE_OF_THE_REST_OF_YOUR_TREE])
```

Submission

Please submit the **AAgent_Python** folder compressed in a single zip file with all the files you consider necessary to test your implementation.

Include a *PDF* document where you explain how you implemented the behaviour trees and highlight the main features of your implementation (4 pages maximum). It has to be informative, so avoid any redundant information (I know what a Behavior Tree is and I know about asynchronous programming, don't waste your time explaining things like that, only what you want me to notice about your implementation). Also include instructions for testing your agents if necessary.